

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 10

[illegible]

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_1	D	S_2
cont.											$-\frac{2h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}}{\sqrt{3}\epsilon}2\lambda_{S_2}^{(2)}$
											$-\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2}}{\sqrt{3}}2\lambda_{S_2}^{(2)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$			$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$		$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$			$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$-\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$			$-\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$			$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$		$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$			$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_1}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}S_{1,j_1}^\dagger S_{1,j_2}^\dagger$			$+h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_1}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{2,j_1}^\dagger S_{2,j_2}^\dagger$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$			$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$		$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$			$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$			$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{1}{2}C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$		$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\frac{1}{2}C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
								$-\frac{C_{S_1^\dagger S_1}h\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$			$-\frac{C_{S_2^\dagger S_2}h\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
								$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$
								$\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}}{\epsilon}$			$\frac{h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}1\lambda_{S_1}^{(1)}$			$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}1\lambda_{S_2}^{(1)}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 \ D$	S_2
cont.									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$		$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$
									$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
cont.									$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$	$-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
									$-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
cont.									$-\frac{4\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$	$+\frac{1}{2}S_2^2\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
									$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
cont.									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$	$-\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
cont.									$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}$	
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	
									$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1		$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
									$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$	$-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$
cont.									$-\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$
									$-\frac{4\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$	$+\frac{1}{2}S_2^2\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
cont.									$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$	$-\hbar\overline{y_{2,i_1,r_1}}S_{2,j_2}^2y_{2,i_1,r_1}\left(S_{2,j_1}^\dagger\right)^2\lambda_{S_2}^{(1)}$
									$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}$	
cont.									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_3,j_1,j_4,j_2}}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$		$+\frac{2\hbar\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{\epsilon}$
									$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$
cont.									$-\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\sqrt{3}}$	$-C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
										$-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_2} y_{2,i_1,r_1} S_{2,j_2}^\dagger}{\epsilon}$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}$ $\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.										$-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_2^\dagger\right)^2 \lambda_{S_2}^{(1)}}{\epsilon}$
										$+\frac{2h\bar{y}_{1,i_1,r_1} y_{1,i_2,r_2} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_1^\dagger S_1} \overline{h\bar{y}_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{1,i_1,r_1} \overline{y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}$ $-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} {}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$		$+D_{\mu_1} \left(S_{1,j_1}\right)^\dagger D_{\mu_1} \left(S_{1,j_1}\right)$	$+\frac{2h\bar{y}_{2,i_1,r_1} y_{2,i_2,r_2} S_{2,j_2}^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger\right)^2}{3\epsilon}$ $+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2} \overline{h\bar{y}_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{4h\bar{y}_{2,i_1,r_1} \overline{y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} {}^1\lambda_{S_2}^{(1)}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\d$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
cont.									$\frac{2h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}\epsilon}$	$\frac{2h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$
									$-\frac{h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
									$-\frac{h\overline{y}_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_1,j_4,j_2} 2\lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_3,j_1,j_4,j_2} 2\lambda_{S_2}^{(2)}}{\sqrt{3}}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{27\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar(W_{\mu_1\mu_2}^{A_1})^2}{\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$			$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar(W_{\mu_1\mu_2}^{A_1})^2}{2\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}$	$(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{9\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_2} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}$
										$-\frac{\hbar y_{2,i_1,r_1} S_1 S_2 j_2 y_{2,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} \epsilon M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger}{\epsilon}$	$-\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_2 j_1 y_{1,i_2,r_1} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3} \epsilon}$
								$+\frac{2\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1} \dagger S_1 \dagger$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_2,r_2} y_{2,i_1,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} \epsilon M_F}$	$+H_{a_2} S_1 H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar y_{1,i_1,r_1} y_{2,i_2,r_1} S_1 S_2 j_1 y_{1,i_1,r_2} y_{2,i_2,r_2} S_1 \dagger S_{2,j_1} \dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1} \dagger H_{a_2} \dagger}{\sqrt{3} M_F}$	$-\frac{2\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$-\frac{2\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
									$-\hbar y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1} \dagger S_1 \dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{\hbar y_{1,i_1,r_1} S_1 S_2 j_2 y_{1,i_1,r_1} S_1 \dagger S_{2,j_1} \dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3} \epsilon}</$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	S_1	H	S_2
cont.											$-\frac{2h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
											$-\frac{h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}$	$-\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+\frac{4h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $+S_1 S_{2,j_2} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}$	
								$+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-h\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}$	$-\frac{2h\bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{h\bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{2h\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}}{\sqrt{3}}$	
cont.											
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1}{}^* H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1}{}^* H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}$	$+\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$	
								$+\frac{2h\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{h\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda_{H1}^{(1)}$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2 \lambda_{H1}^{(2)}$	$-\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda_{12}^{(1)}$ $-\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} L_{a_4, i_2}{}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} \epsilon M_F}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} L_{a_4, i_2}{}^T C P_L L_{a_2, i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1, a_2} \epsilon_{a_3, a_4}}{\sqrt{3} M_F}$ $+\frac{2h\overline{\lambda_{T3} L_{a_3, i_1}} C P_R \overline{L_{a_4, i_2}}{}^T y_1, i_1, r_1 y_2, i_2, r_1 \epsilon_{a_1, a_3} \epsilon_{a_2, a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} \epsilon M_F}$	$+\frac{2h\overline{y_2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_1} S_{1, j_2} y_1, i_1, r_1 \lambda_{T3} \overline{I_{y_2 a_1, j_1}} \overline{I_{y_2 a_2, j_2}}}{\sqrt{3} \epsilon}$ $+\frac{h\overline{y_2, i_1, r_1} H_{a_1} H_{a_2} S_{1, j_1} S_{1, j_2} y_1, i_1, r_1 \lambda_{T3} \overline{I_{y_2 a_1, j_1}} \overline{I_{y_2 a_2, j_2}}}{\sqrt{3}}$ $-\frac{2h\overline{y_1, i_1, r_1} Y_{\epsilon, i_2, i_3} H_{a_2} S_{1, j_2} y_1, i_2, r_1 Y_{\epsilon, i_1, i_3} \overline{I_{y_1, a_1, j_2}} \overline{I_{y_1 a_2, j_1}} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1} Y_{\epsilon, i_2, i_3} H_{a_2} S_{1, j_2} y_1, i_2, r_1 Y_{\epsilon, i_1, i_3} \overline{I_{y_1, a_1, j_2}} \overline{I_{y_1 a_2, j_1}} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_1, i_1, r_1} \overline{\lambda_{T3} y_2, i_1, r_1} \overline{I_{y_2, a_1, j_2}} \overline{I_{y_2, a_2, j_1}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{\sqrt{3} \epsilon}$ $+\frac{h\overline{y_1, i_1, r_1} \overline{\lambda_{T3} y_2, i_1, r_1} \overline{I_{y_2, a_1, j_2}} \overline{I_{y_2, a_2, j_1}} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1, j_1}{}^\dagger S_{1, j_2}{}^\dagger}{\sqrt{3}}$ $+\frac{H_{a_2} S_{1, j_2} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\sqrt{3}}$ $-\frac{h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2} h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^1 \lambda_{H1}^{(1)}}{\epsilon}$ $+\frac{H_{a_2} S_{1, j_2} H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2} h\overline{y_1, i_1, r_1} H_{a_2} S_{1, j_2} y_1, i_1, r_1 H_{a_1}{}^\dagger S_{1, j_1}{}^\dagger \mathcal{I}_{H1, a_1, a_2, j_1, j_2}{}^2 \lambda_{H1}^{(2)}}{\epsilon}$	$+C \text{NewScalar1} \text{NewScalar2} h S_{1, j_1} S_{2, j_2} \text{Bar}(\text{cg1})_{j_1, j_2}$ $+\frac{1}{2} C_{S_1^\dagger S_2} h \overline{y_2, i_1, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_1 \text{Bar}(\text{cg1})_{j_1, j_2}$ $+\frac{C_{S_2^\dagger S_2} h \overline{y_2, i_1, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_1 \text{Bar}(\text{cg1})_{j_1, j_2}}{\epsilon}$ $-\frac{h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} S_{2, j_4} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_1, j_2} \text{Bar}(\text{cg1})_{j_3, j_4}}{\epsilon}$ $+h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} S_{2, j_4} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_1, j_2} \text{Bar}(\text{cg1})_{j_3, j_4}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{1, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{1, j_2} S_{2, j_2} y_1, i_1, r_2 y_1, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{1, j_1}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{2, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{2, j_1}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_2, i_1, r_1 y_2, i_2, r_2} S_{1, j_2} S_{2, j_1} S_{2, j_3} y_1, i_1, r_2 y_2, i_2, r_2 \text{Bar}(\text{cg1})_{j_2, j_3} S_{2, j_1}{}^\dagger}{\epsilon}$ $+\frac{1}{2} C_{S_2^\dagger S_2} h \overline{y_1, i_1, r_1} \text{cg1}_{j_1, j_2} y_2, i_1, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+\frac{C_{S_1^\dagger S_2} h \overline{y_1, i_1, r_1} \text{cg1}_{j_1, j_2} y_2, i_1, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_2} S_{2, j_1} y_1, i_2, r_1 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_1, i_2, r_1 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $-2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_2} S_{2, j_1} y_1, i_2, r_2 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_1} S_{2, j_2} y_1, i_2, r_2 y_2, i_1, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} S_{1, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_1 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger}{\epsilon}$ $+2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} S_{1, j_1} S_{2, j_2} y_1, i_1, r_2 y_2, i_2, r_2 S_{1, j_1}{}^\dagger S_{2, j_2}{}^\dagger$ $+\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_2} \text{cg1}_{j_1, j_3} S_{2, j_2} y_2, i_1, r_2 y_2, i_2, r_1 S_{1, j_1}{}^\dagger S_{2, j_3}{}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_1, i_1, r_1 y_2, i_2, r_1} \text{cg1}_{j_2, j_3} S_{1, j_1} y_1, i_1, r_2 y_2, i_2, r_2 S_{1, j_1}{}^\dagger S_{2, j_3}{}^\dagger}{\epsilon}$ $+\frac{2h\overline$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger \lambda_{S_2}^{(1)}$ $+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}}{\epsilon}$
cont.													$+\frac{1}{2}h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{S_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}{}^*H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1},j_2}\mathcal{I}_{y_{1,a_2},j_1}{}^*H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$			
cont.								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$				
cont.								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$				
cont.													$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$
													$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$
													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.												$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}2\lambda_{12}^{(2)}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^TC P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$		$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{1,a_1,j_2}}\mathcal{I}_{y_{1,a_2,j_1}}H_{a_1}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}}{\epsilon}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $-2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.								$+\frac{2h\lambda_{T3}L_{a_3,i_1}C P_R \overline{L_{a_4,i_2}}{}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}1\lambda_{H1}^{(1)}$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}}{\epsilon}$	$+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$		
cont.										$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$ $-\frac{1}{2}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}2\lambda_{H1}^{(2)}$	$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}1\lambda_{12}^{(1)}$ $-\sqrt{3}S_{1,j_2}S_{2,j_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3}S_{1,j_1}S_{2,j_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$ $+\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$		
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}$ $+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$	
cont.												$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
cont.													$+ \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$+ \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}$
													$- \frac{1}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{1}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{2h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
													$- \frac{10h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$
cont.													$- \frac{10}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{8}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
													$+ \frac{2}{3} \frac{h\bar{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
													$- \frac{1}{2} \frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}}{\epsilon}$
cont.													$+ \frac{3h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{1,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
cont.													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$+ \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$
													$- \frac{3}{2} \frac{h\bar{y}_{2,i_1,r_1}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\bar{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\bar{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}^2 \lambda_{12}^{(2)}$
cont.										+CNewScalar1NewScalar2hS1,j1S2,j2Bar(cg3)j1,j2 +12CS2†S2h̄y2,i1,r1S1,j1S2,j2y1,i1,r1Bar(cg3)j1,j2 +CS2†S2h̄y2,i1,r1S1,j1S2,j2y1,i1,r1Bar(cg3)j1,j2 +ϵ 5h̄y2,i1,r1y2,i2,r2S1,j1S1,j3S2,j2S2,j4y1,i1,r2y1,i2,r1Bar(cg3)j1,j2Bar(cg3)j2,j4 9ϵ 19h̄y2,i1,r1y2,i2,r2S1,j1S1,j3S2,j2S2,j4y1,i1,r2y1,i2,r2Bar(cg3)j1,j2Bar(cg3)j3,j4 +10h̄y1,i1,r1y2,i2,r2S1,j1S1,j3S2,j2y1,i1,r2y1,i2,r1Bar(cg3)j1,j2S1,j1† 3√3ϵ 2h̄y1,i1,r1y2,i2,r1S1,j1S1,j2S2,j3y1,i1,r2y1,i2,r2Bar(cg3)j2,j3S1,j1† 3√3 2h̄y1,i1,r1Ȳe,i2,i3H̄a2S1,j2y1,i2,r1Ȳe,i1,i3H̄a1†S1,j1†IH1,a1,a2,j1,j21 +10h̄y2,i1,r1y2,i2,r2S1,j2S2,j1S2,j3y1,i1,r2y2,i2,r1Bar(cg3)j2,j3S2,j1† 3√3ϵ 2h̄y2,i1,r1y2,i2,r1S1,j2S2,j1S2,j3y1,i1,r2y2,i2,r2Bar(cg3)j2,j3S2,j1† 3√3 +hCNewScalar1NewScalar2cg3j1,j2S1,j1†S2,j2† +12CS2†S2h̄y1,i1,r1cg3j1,j2y2,i1,r1S1,j1†S2,j2† +CS2†S2h̄y1,i1,r1cg3j1,j2y2,i1,r1S1,j1†S2,j2† +2h̄y1,i1,r1y2,i2,r2S1,j2S2,j1y1,i2,r1y2,i1,r2S1,j1†S2,j2† 3ϵ 10h̄y1,i1,r1y2,i2,r2S1,j1S2,j2y1,i1,r2y2,i2,r1S1,j1†S2,j2† 3ϵ 10h̄y1,i1,r1y2,i2,r2S1,j1S2,j2y1,i1,r2y2,i2,r1S1,j1†S2,j2† 3ϵ +8h̄y1,i1,r1y2,i2,r2S1,j2S2,j1y1,i1,r2y2,i2,r2S1,j1†S2,j2† 3ϵ +8h̄y1,i1,r1y2,i2,r2S1,j2S2,j

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{3h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{3}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$ $+\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$ $+\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$
cont.													$-\frac{3h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{3}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.													$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$ $-\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
cont.													$-\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1			$-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$ $-\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}{}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2h\lambda_{T3}\overline{L_{a_3,i_1}}CP_R\overline{L_{a_4,i_2}}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}\epsilon}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{H1}^{(1)}$ $+\frac{H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H1}^{(2)}$	$+\frac{2h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $+\frac{10h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon}$ $-\frac{10}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{8}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $+\frac{2}{3}\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0			$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
										$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}\epsilon M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-\frac{2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}{\epsilon}{}^1$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+S_2S_{1,j_2}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$	
cont.									$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$		
cont.										$-\frac{h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$		
T3-D	0	3	0	1	1	2	$\frac{1}{2}$			$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
										$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}\epsilon M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-\frac{2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}{\epsilon}{}^1$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+S_2S_{1,j_2}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$	
cont.									$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$		
cont.										$-\frac{h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$		
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1			$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
										$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}\epsilon M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-\frac{2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}{\epsilon}{}^1$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+S_2S_{1,j_2}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$	
cont.									$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$		
cont.										$-\frac{h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$		
T3-D	2	3	1	1	2	2	$\frac{3}{2}$			$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$	
										$+\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}$	$-\frac{2h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
										$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}\epsilon M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-\frac{2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}{\epsilon}{}^1$	$+\frac{4h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $+S_2S_{1,j_2}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$	
cont.									$+\frac{2h\lambda_{T3}L_{a_3,i_1}}{\sqrt{3}M_F}C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger$	$-2i\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{Y}_{\epsilon,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2}S_{1,j_2}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}$ $-\frac{2h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{2h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y}_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$		
cont.										$-\frac{h\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{2h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}}{\epsilon}$ $-h\overline{y}_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1\lambda_{12}^{(1)}$		

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	S_2
cont.										$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$	
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	
										$+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$	
cont.										$+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	
										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$	
										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	
cont.										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}$	
										$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon}$	
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	
cont.										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon}$	
										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	
										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0			$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
										$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}CPL_{a_4,i_2}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{-1}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{-1}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a_3,i_1}CPL_{a_4,i_2}^Ty_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$-\frac{4h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon}$
cont.										$-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2\lambda_{H1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$
										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
cont.										$\begin{aligned} & +S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)} \\ & -\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & -\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & -S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)} \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)} \\ & +\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & +\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
cont.										$\begin{aligned} & -\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \\ & +\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & +\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & +\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & +\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & -\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2} {}^2\lambda_{12}^{(2)}}{\sqrt{3}} \end{aligned}$
cont.										$\begin{aligned} & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)} \\ & -\frac{2h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & -\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$
										$\begin{aligned} & -\frac{2h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}\epsilon} \\ & -\frac{h\bar{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2} {}^3\lambda_{12}^{(3)}}{\sqrt{3}} \end{aligned}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & -\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F} \\ & -\frac{2\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F} \\ & -\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F} \end{aligned}$	$\begin{aligned} & -\frac{2h\bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon} \\ & -\frac{2h\bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2i\sqrt{\frac{2}{3}}h\bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^\dagger}{\epsilon} \end{aligned}$	$\begin{aligned} & -\frac{4h\bar{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j$

[illegible]

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}$
cont.													$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}e}$
cont.													$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_3,j_1,j_4,j_2}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 \ell$	$S_2 H$
cont.										$-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2\lambda_{H2}^{(2)}}{\sqrt{3}}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}{}^\dagger\lambda_{S_1}^{(1)}}$		$-\frac{2h\overline{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\epsilon}$ $-2h\overline{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y2,a_1,j_1}\mathcal{I}_{y2,a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(1)}$ $-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(1)}}{\epsilon}$ $-\frac{1}{2}h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(2)}$ $-\frac{h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(2)}}{\epsilon}$ $-\frac{1}{2}h\overline{y}_{2,i_1,r_1}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^\dagger\lambda_{H2}^{(2)}$
cont.										
cont.										
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-C_{S_1^\dagger S_1}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}{}^\dagger\lambda_{S_1}^{(1)}}$	$\frac{3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}g_2^2}{8eM_F}$ $+\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}g_2^2}{8M_F}$ $+3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}L_{a_2,i_2}}{}^T CPL_{a_1,i_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2$ $+\frac{8eM_F}{8eM_F}$ $+h\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}L_{a_2,i_2}}{}^T CPL_{a_1,i_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2$ $+\frac{8M_F}{8M_F}$ $+3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}$ $+\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{2M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{2eM_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{4M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{4eM_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{4eM_F}$ $-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{8M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{eM_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y1,a_1,j_1}\mathcal{I}_{y1,a_2,j_2}}{2M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_1}}\overline{L_{a_2,i_2}}{}^T CPL_{a_$	

[illegible]

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 \ell$	$S_2 H$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\sqrt{3}C_{S_1^\dagger S_1} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger \\ -\frac{5}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ -\sqrt{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_4,j_2}^{(1)} \lambda_{S_1}^{(1)}$	$-\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon} \\ -\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ -\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{(1)}}{\sqrt{3}\epsilon} \\ -\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^{(1)}}{\sqrt{3}} \\ +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)} \lambda_{H2}^{(1)} \\ -\frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)} \lambda_{H2}^{(1)}}{\epsilon} \\ -\frac{1}{2} \sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(1)} \lambda_{H2}^{(1)} \\ +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)} \lambda_{H2}^{(2)} \\ -\frac{\sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)} \lambda_{H2}^{(2)}}{\epsilon} \\ -\frac{1}{2} \sqrt{3} h \bar{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^{(2)} \lambda_{H2}^{(2)}}{\epsilon}$	
cont.										
cont.										
cont.										
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2 \\ -\frac{2C_{S_1^\dagger S_1} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$	$\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
cont.								$-\frac{4}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ 2h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_4,j_2}^{(1)} \lambda_{S_1}^{(1)} \\ -\frac{2h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_4,j_2}^{(2)} \lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger \\ +H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)} \\ \frac{2h\bar{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)}}{\epsilon} \\ -h\bar{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)}$	
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2 \\ -\frac{2C_{S_1^\dagger S_1} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$	$\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon}$	
cont.								$-\frac{4}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ 2h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_4,j_2}^{(1)} \lambda_{S_1}^{(1)} \\ -\frac{2h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_3,j_4,j_2}^{(2)} \lambda_{S_1}^{(2)}}{\sqrt{3}}$	$-2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger \\ +H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)} \\ \frac{2h\bar{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)}}{\epsilon} \\ -h\bar{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^{(1)} \lambda_{H2}^{(1)}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2}{3} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger\right)^2 \\ -\frac{2C_{S_1^\dagger S_1} h \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\sqrt{3}}$	$\frac{2h\bar{y}_{2,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_1	ℓ	S_2	H
$cont.$													$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger T_{H^2,a_1,a_2,j_2,j_1}}{\sqrt{3}}{}^2\lambda_{H^2}^{(2)}$

Field combinations 5 of 10

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 F$	$S_2 \ell$	$F B_{\mu\nu}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger \\ & -\frac{5}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S_2,j_1,j_3,j_2,j_4}^{(1)} \lambda_{S_2}^{(1)} \end{aligned}$	$\begin{aligned} & + \frac{3h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2} g_2^2}{8 M_F} \\ & + \frac{h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2} g_2^2}{8 M_F} \\ & + \frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c g 1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8 M_F} \\ & + \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c g 1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_2^2}{8 M_F} \\ & + \frac{3h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_3,i_3} g_1^2 S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & + \frac{h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & + \frac{\overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{2 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & - \frac{3 \sqrt{3} h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & - \frac{3 \sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{8 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{2 M_F} \\ & - \frac{\sqrt{3} h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \text{Bar}(cg1)_{a_1,j_1,a_2,j_2}}{4 M_F} \\ & + \frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c g 1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8 M_F} \\ & + \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c g 1_{a_1,j_1,a_2,j_2} g_1^2 y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8 M_F} \\ & + \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T c g 1_{a_1,j_1,a_2,j_2} y_{2,i_1,r_1} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2 M_F} \end{aligned}$	
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	S_2	ℓ	F	$B_{\mu\nu}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger}$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1}}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger}$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1}}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger}$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1}}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_2^\dagger \right)^2$					$-\frac{9}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_2} y_{2,i_1,r_1} S_2^\dagger}$					
								$-2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1}}} \varsigma_2^2 y_{2,i_1,r_1} \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$					
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}}}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}$					
								$-\frac{2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}}{\sqrt{3}}$					
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}}}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}$					
								$-\frac{2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}}{\sqrt{3}}$					
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} \varsigma_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2$					$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2$
								$-\frac{2C_{S_2^\dagger S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}}}{\sqrt{3}}$					
								$-\frac{\frac{4}{3} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2}}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger + 2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(1)}}}{\sqrt{3}}$					
cont.								$2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}$					
								$-\frac{2\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger} I_{S_{2,j_3,j_1,j_4,j_2}} \lambda_{S_2}^{(2)}}}{\sqrt{3}}$					

[illegible]

Field combinations 6 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
<i>cont.</i>										$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{2\sqrt{3}}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda T_3} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$
									$+\frac{3}{8} \frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\epsilon}$
									$+\frac{\overline{\sqrt{3} h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{2} \frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\sqrt{3}\epsilon}$
<i>cont.</i>									$+\frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{2\sqrt{3}}$
<i>cont.</i>									$+\frac{3}{8} \frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1}} \overline{H_{a_2}} S_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger$
									$+\frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{2} \frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\sqrt{3}\epsilon}$
<i>cont.</i>										$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{2\sqrt{3}}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2$	$+\frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda T_3} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$
									$+\frac{3}{8} \frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\epsilon}$
									$+\frac{\overline{\sqrt{3} h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{2} \frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{\sqrt{3}\epsilon}$
<i>cont.</i>									$+\frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger}{2\sqrt{3}}$
<i>cont.</i>									$+\frac{3}{8} \frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1}} \overline{H_{a_2}} S_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger$
									$+\frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{2} \frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\epsilon}$
									$+\frac{3}{8} \frac{\overline{\sqrt{3} h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{\sqrt{3}\epsilon}$
<i>cont.</i>										$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} \lambda T_{3,S_2,j_1}^\dagger}{2\sqrt{3}}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$		$+\frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda T_3} S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8} \frac{\overline{h y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\overline{h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4} \frac{\overline{h y_{1,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\overline{h y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
<i>cont.</i>									$+\frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \frac{\overline{h y_{2,i_1,r_1}} \overline{\lambda T_3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
<i>cont.</i>									$+\frac{3}{8} \frac{\overline{h y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$+\overline{H_{a_1}} \overline{H_{a_2}} S_1 \mathcal{I} T_{3,a_1,a_2,j_2} \lambda T_{3,S_2,j_1}^\dagger$
									$+\frac{\overline{h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda T_{3,S_2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\overline{h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4} \frac{\overline{h y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda T_{3,S_2,j_1}^\dagger}{2\epsilon}$
									$+\frac{3}{8} \frac{\overline{h y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda T_{3,S_2,j_1}^\dagger}{2\epsilon}$
<i>cont.</i>										$-\frac{1}{4} \frac{\overline{h y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} \lambda T_{3,S_2,j_1}^\dagger}{2\epsilon}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	$S_1 S_2 H$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
										$-\frac{\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3S_{2,j_2}}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar y_{1,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\hbar y_{2,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
										$-\frac{1}{4}\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3S_{2,j_2}}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$		$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar y_{1,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar y_{2,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
										$-\frac{1}{4}\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\lambda\overline{T_3S_{2,j_2}}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
									$+\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda T_3S_{2,j_2}}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar y_{2,i_1,r_1}\overline{\lambda T_3S_{2,j_2}}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
									$+\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$
cont.									$+\frac{3}{8}\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
										$-\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	$S_1 S_2 H$
<i>cont.</i>										$-\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$\begin{aligned} & +\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\ & +\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1} \\ & +\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\ & +\frac{3}{8}\sqrt{3h\bar{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1} \\ & +\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\ & +\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger \\ & +\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger \\ & +\frac{3}{8}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger \end{aligned}$	$\begin{aligned} & -\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon} \\ & -\frac{2h\bar{y}_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\text{Bar}(\text{cg}4)_{a_2,j_1,a_1,j_2}H_{a_1}^\dagger}{\epsilon+\lambda T_3S_{2,j_2}T_3^*_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \end{aligned}$
<i>cont.</i>										$\begin{aligned} & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \end{aligned}$
<i>cont.</i>										$\begin{aligned} & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \end{aligned}$
<i>cont.</i>										$\begin{aligned} & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \end{aligned}$
<i>cont.</i>										$\begin{aligned} & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger}{2\epsilon} \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \end{aligned}$
<i>cont.</i>										$\begin{aligned} & -\frac{1}{4}\sqrt{3h\bar{y}_{2,i_1,r_1}}\overline{\lambda T_3}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}\lambda T_3S_{2,j_1}^\dagger \\ & -\frac{\sqrt{3h\bar{y}_{1,i_1$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	S_1S_2H
cont.									$+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $+\frac{\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}}{4\epsilon}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
										$-\frac{1}{2}\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\sqrt{3}\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{3}{8}\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}}{4\epsilon}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H\ell e$	$S_1 S_2 H$
cont.									$+\frac{3}{8}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\epsilon}$ $-\frac{1}{2}\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{1}{2}\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_2y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$+\overline{\lambda_{T3}}S_{2,j_2}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3}^*S_{a_1,a_2,j_1,j_2}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$ $+\frac{H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\overline{hy_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{hy_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{1,i_1,r_1}}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{hy_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{hy_{2,i_1$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW\mu\nu$	$H\ell e$	S_1S_2H
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger$
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_{T3}}S_{2,j_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger$
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_{T3}}S_{2,j_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_1,j_2}}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger$
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{1,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}}\overline{Y}_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$	$-\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{\mathcal{I}T_{3,a_1,a_2,j_2,j_1}}\lambda_{T3}S_{2,j_1}^\dagger}{2\sqrt{3}}$

Field combinations 7 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger} \mathcal{I}_{12,j_2,j_1}^{-1} \lambda_{12}^{(1)}$	$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}}{M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$	$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger} S_1^\dagger \mathcal{I} H_{1,a_1,a_2}^{-1} \lambda_{H1}^{(1)}$
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{2M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{3\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{4\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}}{8M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}}{\sqrt{3}\epsilon M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} \overline{y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}}{2\sqrt{3}M_F} \mathcal{C} P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $+\frac{\overline{L_{a_1,i_1}}}{M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\sqrt{3}\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{8M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{2M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{\sqrt{3}\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{\hbar \overline{y_{2,i_3,r_1} L_{a_1,i_1}}}{2\sqrt{3}M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$		
cont.								$-\frac{\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{4\epsilon M_F} \mathcal{C} P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger$ $-\frac{3\hbar \overline{y_{1,i_3,r_1} L_{a_1,i_1}}}{8M_F} \$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	$S_1 S_2 \ell$	$S_1 F H$
cont.											$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
cont.											$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
cont.											$\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
cont.											$\frac{3\sqrt{3}h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
cont.											$\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}L_{a_2,i_2}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
cont.											$\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}{2\sqrt{3}M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_1^\dagger$	
											$\frac{h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$	
cont.											$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}}{8M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$	
											$+\frac{\overline{L}_{a_1,i_1}}{M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$	
cont.											$\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}}{4\epsilon M_F}{}^T C P_R \overline{L}_{a_2,i_2}{}^T S_1 y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\overline{I}T_{3,a_1,a_2,j_1}S_{2,j_1}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{y}_{2,i_3,r_2}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{2\sqrt{3} M_F}$	
									$-\frac{h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_2,r_1} \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_2,r_1} \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{\frac{8M_F}{M_F} + \frac{\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}}$	
									$-\frac{\sqrt{3}h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{1,i_3,r_1} \bar{y}_{2,i_2,r_2} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{1,i_3,r_1} \bar{y}_{2,i_2,r_2} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{2M_F}$	
									$-\frac{h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_2} \bar{y}_{2,i_3,r_1} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_2} \bar{y}_{2,i_3,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{2\sqrt{3} M_F}$	
									$-\frac{h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{1,i_2,r_1} \bar{y}_{2,i_3,r_2} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{1,i_2,r_1} \bar{y}_{2,i_3,r_2} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{\frac{8M_F}{M_F} + \frac{\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_1} \mathcal{I} F_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}}$	
									$-\frac{\sqrt{3}h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{y}_{2,i_2,r_2} S_1 S_{2,j_1} \bar{y}_{1,i_2,r_1} \bar{y}_{2,i_1,r_2} S_1^\dagger \lambda_{12}^{(1)}}{\frac{8M_F}{M_F} + \frac{\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_2} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger \lambda_{12}^{(1)}}{M_F}}$	
cont.									$-\frac{h\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$-\frac{h\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{4h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_2} S_1 S_{2,j_1} \bar{y}_{1,i_2,r_1} \bar{y}_{2,i_1,r_2} S_1^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$ $-2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} \bar{y}_{1,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}$ $+2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} \bar{y}_{2,i_1,r_1} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}{}^1 \lambda_{12}^{(1)}$	$-\frac{9h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$ $-\frac{3h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4M_F}$ $+\frac{\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{M_F}$	$-2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{Y}_{e,i_2,i_3} H_{a_1} S_1 \bar{y}_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger$ $-2h \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} H_{a_2} S_1 \bar{y}_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}{}^1 \lambda_{H1}^{(1)}$
cont.									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_3,r_2}\bar{y}_{2,i_2,r_2}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_3,r_2}\bar{y}_{2,i_2,r_2}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_3,r_2}\bar{y}_{2,i_2,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{\epsilon M_F}$	
									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_3,r_2}\bar{y}_{2,i_2,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{2M_F}$	
cont.									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\bar{y}_{1,i_3,r_1}\bar{y}_{2,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
									$-\frac{3\sqrt{3}h\bar{y}_{1,i_3,r_1}\bar{y}_{2,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{8M_F}$	
cont.									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{\epsilon M_F}$	
									$-\frac{h\bar{y}_{1,i_1,r_1}\bar{y}_{1,i_2,r_2}\bar{y}_{2,i_3,r_1}\bar{L}_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \bar{y}_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{2\sqrt{3} M_F}$	
cont.									$-\frac{h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_2,r_1} \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_2,r_1} \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{8M_F}$	
cont.									$-\frac{9h\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T g_Y^2 S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T g_Y^2 S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$	
cont.									$+\frac{\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$	
									$-\frac{\sqrt{3}h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_1} C P_R \bar{L}_{a_2,i_2}{}^T S_1 \bar{y}_{1,i_3,r_2} \bar{y}_{2,i_1,r_2} \bar{y}_{2,i_2,r_1} \mathcal{I} T_{3a_1,a_2,j_1} S_{2,j_1}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$\frac{3\sqrt{3}\overline{h\overline{y_2}}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_1}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\epsilon M_F}$			
cont.													$\frac{h\overline{y_1}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2M_F}$			
													$\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$			
cont.													$\frac{h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{2\sqrt{3}M_F}$			
													$\frac{h\overline{y_1}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$\frac{3h\overline{y_1}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$\frac{\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$\frac{3\sqrt{3}h\overline{y_2}_{i_3,r_1}L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_1}_{i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$\frac{h\overline{y_1}_{i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\overline{IT_3}_{a_1,a_2,j_1}S_1^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}^TS_{1y_1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_3}_{a_1,a_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$	
									$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{8M_F}$		
cont.									$-\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{4\epsilon M_F}$		
									$-\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger}{8M_F}$		
									$-\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_1^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$		
cont.									$-\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_1^\dagger \lambda_{12}^{(1)}}{M_F}$		
									$-\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$		
cont.									$-\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{y}_{a_1,a_2,j_1}^* S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$+ \frac{3h\overline{y}_{1,i_2,r_1} \overline{y}_{2,i_1,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I} y_{1,a_1,j_1} \overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$	$-2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	
									$+ \frac{h\overline{y}_{1,i_2,r_1} \overline{y}_{2,i_1,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} g_2^2 S_{2,j_2} \overline{I} y_{1,a_1,j_1} \overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{H}_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T \overline{H}_{1,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{H1}^{(1)}$	
									$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T_{12,j_2,j_3,j_1-j_4} \lambda_{12}^{(1)}$	$+ \frac{\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \overline{I} y_{1,a_1,j_1} \overline{I} y_{2,a_2,j_2}^* S_{1,j_1}^\dagger}{M_F}$	$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T \overline{H}_{1,a_1,a_2,j_1} S_{2,j_1}^\dagger \lambda_{H1}^{(2)}$
									$-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T_{12,j_2,j_3,j_1-j_4} \lambda_{12}^{(1)}$	$\frac{h\overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_2,r_2} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I} y_{a_1,j_1}^* \overline{I} y_{a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	
cont.									$+ \sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{3h\overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_2,r_2} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I} y_{a_1,j_1}^* \overline{I} y_{a_2,j_2}^* S_{1,j_1}^\dagger}{8M_F}$	
									$-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_2,r_2} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I} y_{a_1,j_1}^* \overline{I} y_{a_2,j_2}^* S_{1,j_1}^\dagger}{2\epsilon M_F}$	
									$+\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_2,r_2} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I} y_{a_1,j_1}^* \overline{I} y_{a_2,j_2}^* S_{1,j_1}^\dagger}{4M_F}$	
cont.									$-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda$		

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}\overline{y}_{2,i_3,r_3}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}\overline{y}_{2,i_3,r_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_2}\overline{y}_{2,i_3,r_2}I_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}^*S_{1,j_1}^\dagger}{4M_F}$	
									$+\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4M_F}$	
									$+\frac{3h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$+\frac{h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$+\frac{\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_3,r_1}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{M_F}$	
cont.									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{2\epsilon M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$	
									$\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}I_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F}$	
									$+\frac{h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4M_F$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.										$+\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8M_F}$			
										$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{24M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4M_F}$			
cont.										$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$			$\frac{3h\overline{y_{1,i_1,r_1}}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8M_F}$			
cont.													$\frac{3h\overline{y_{1,i_3,r_1}}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4M_F}$			
cont.													$\frac{3h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{8M_F}$			
													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{24M_F}$			
cont.													$\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}{}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}x_{y_1,a_1,j_1}^*I_{y_2,a_2,j_2}S_{1,j_1}^\dagger}{4M_F}$			
													$\frac{3h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_1}L_{a_2,i_2}{}^Tg_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{46M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{9h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{46M_F}$			
cont.													$\frac{3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
cont.													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}x_{y_1,a_1,j_2}I_{y_2,a_2,j_1}S_{2,j_1}^\dagger}{4M_F}$			
													$\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}}{}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$			
cont.									$\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I_{y_1,a_1,j_2} I_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$ $\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I_{y_1,a_1,j_2} I_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
								$+2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}}} \lambda_{12}^{(1)}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}}} \lambda_{12}^{(1)}$	$+ \frac{3h\overline{y_{1,i_2,r_1} y_{2,i_1,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$ $+ \frac{h\overline{y_{1,i_2,r_1} y_{2,i_1,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4M_F}$ $- \frac{6h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{\epsilon M_F}$	$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} I_{y_1,a_1,j_2} \overline{I_{y_1,a_2,j_1}}^* H_{a_1}^\dagger S_{1,j_1}^\dagger}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}}} \lambda_{12}^{(1)}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger I_{H1,a_1,a_2,j_1,j_2}}} \lambda_{12}^{(2)}$
cont.								$+ \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}} \lambda_{12}^{(2)}$ $- \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}} \lambda_{12}^{(2)}$ $+ \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}} \lambda_{12}^{(2)}$		
cont.								$- \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}} \lambda_{12}^{(2)}$ $-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{1}{\overline{y_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_2,j_3,j_1,j_4}}} \lambda_{12}^{(2)}$	$+ \frac{2h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{M_F}$ $+ \frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{M_F}$ $\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$ $\frac{3h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{2\epsilon M_F}$ $- \frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$ $- \frac{3h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$ $- \frac{3h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_1,a_1,j_1}}^* \overline{I_{y_2,a_2,j_2}}^* S_{1,j_1}^\dagger}{4\epsilon M_F}$ $- \frac{h\overline{y_{1,i_1$	

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+C_{\frac{1}{2}g_2}^2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg3})_{j_1,j_2}} \\ -\frac{5}{9} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_3} S_{2,j_2} S_{2,j_4} y_{1,i_1,r_2} y_{1,i_2,r_1} \text{Bar}(\text{cg3})_{j_1,j_2} \text{Bar}(\text{cg3})_{j_3,j_4}} \\ + \frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{2,j_4}^\dagger}}{3\sqrt{3}} \\ + \frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_2} y_{2,i_2,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{2,j_4}^\dagger}}{3\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger g_2^2}}{8M_F} \\ -\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger g_2^2}}{8M_F} \\ + \frac{9\sqrt{3}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger g_2^2}}{8M_F} \\ + \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger g_2^2}}{8M_F}$	$-2\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,1,r_1} H_{a_1} H_{a_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} \lambda_{T3} \text{Bar}(\text{cg1})_{a_1,j_1,a_2,j_2}} \\ -\frac{4h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
cont.								$+C_{\frac{1}{2}g_2}^2 h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_1,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger} \\ + \frac{2}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger} \\ - \frac{10}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger} \\ + \frac{8}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger} \\ - \frac{10}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger} \\ + \frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}} \\ + \frac{10h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \text{cg3}_{j_1,j_2} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{3\sqrt{3}}$	$-\frac{3h\overline{L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} S_{2,j_1}^\dagger g_2^2}}{8M_F} \\ -\frac{h\overline{L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_2} S_{2,j_1}^\dagger g_2^2}}{8M_F} \\ + \frac{9\sqrt{3}h\overline{L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger g_2^2}}{8M_F} \\ + \frac{3\sqrt{3}h\overline{L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger g_2^2}}{8M_F}$	$-\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,a_1,a_2,j_1,j_2}^{-1} \lambda_{[1]}^{(1)}} \\ -\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,a_1,a_2,j_1,j_2}^{-1} \lambda_{[2]}^{(1)}}$
cont.								$-\frac{5}{9} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} \text{cg3}_{j_3,j_4} \text{cg3}_{j_2,j_4} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_4}^\dagger} \\ -\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_4}^{-1} \lambda_{12}^{(1)}} \\ -\sqrt{3}h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_4}^{-1} \lambda_{12}^{(1)}} \\ + \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_1} \text{Bar}(\text{cg3})_{j_2,j_3} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}} \\ + 3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}} \\ - 3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}} \\ + 3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}} \\ - 3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}} \\ + 2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}} \\ + \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg3}_{j_2,j_3} S_{1,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}}$	$-\frac{h\$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{2}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}$ $-\frac{10}{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}$ $-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(1)}}}$ $-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(1)}}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $-\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}}{2M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	$-\frac{4h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}}$ $+\frac{2h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(1)}}}$ $-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(1)}}}$
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_1,j_2}^{(2)}}}$
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $+\frac{3h\overline{y_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}{M_F^2}$ $-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}^{(2)}}}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	
cont.								$-\sqrt{3}h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\frac{1}{\overline{y_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger}\lambda_{12}^{(2)}}}$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}}{16M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$	
cont.								$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{4M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger$ $+\frac{9\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}}{8M_F}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2$	

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$-\frac{\sqrt{3}L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{2M_F} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$-\frac{3h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8\epsilon M_F} T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \overline{y}_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$\frac{9h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{16M_F} T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \overline{y}_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{3h\overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{4\epsilon M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} \overline{y}_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+\frac{3h\overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} \overline{y}_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{3h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{3h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+\frac{3h\overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8\epsilon M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} \overline{y}_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{9h\overline{y}_{1,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{16M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} \overline{y}_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{3h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{8\epsilon M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+\frac{9h\overline{y}_{2,i_3,r_1} L_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{16M_F} T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}$	
									$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger} \lambda_{12}^{(1)}$	
cont.									$+\frac{\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\epsilon M_F} T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$	
									$+\frac{\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{M_F} T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$	
									$+\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$	
cont.									$+\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{\sqrt{3}\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger \lambda_{12}^{(2)}$	
									$+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\sqrt{3}M_F} T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T 3.a_1.a_2.j_1.j_2 S_{2,j_1}^\dagger \lambda_{12}^{(2)}$	
cont.									$+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\sqrt{3}\epsilon M_F} T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T 3.a_1.a_2.j_1.j_2 S_{2,j_1}^\dagger \lambda_{12}^{(2)}$	
									$+\frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}}{\sqrt{3}M_F} T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T 3.a_1.a_2.j_1.j_2 S_{2,j_1}^\dagger \lambda_{12}^{(2)}$	
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{2}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{3h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$-\frac{4h \log \left(\frac{\epsilon^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{e,i_2,r_2} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} \overline{y}_{e,i_1,r_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$-\frac{10}{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{8M_F} T C P_L L_{a_1,i_1} g_S^2 S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{e,i_2,r_2} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} \overline{y}_{e,i_1,r_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T 3.a_1.a_2.j_1.j_2^{-1}}{\sqrt{3}}$
								$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T 2.j_2.j_3.j_1.j_4^{-1} \lambda_{12}^{(1)}$	$+\frac{3h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{\epsilon M_F} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T 3.a_1.a_2.j_1.j_2^{-1} \lambda_{12}^{(1)}$
								$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T 2.j_2.j_3.j_1.j_4^{-1} \lambda_{12}^{(1)}$	$+\frac{h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{M_F} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
cont.								$+3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$-\frac{\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{2M_F} T C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{I} T 3.a_1.a_2.j_1.j_2^{-2} \lambda_{12}^{(2)}$
								$-3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_3,r_2} \overline{y}_{2,i_2,r_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$+\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_3,r_2} \overline{y}_{2,i_2,r_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
cont.								$-3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{12}^{(2)}$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_3,r_2} \overline{y}_{2,i_2,r_2}}{4\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T 2.j_2.j_3.j_1.j_4^{-2} \lambda_{12}^{(2)}$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_3,r_2} \overline{y}_{2,i_2,r_2}}{8M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
cont.								$-\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \overline{I} T 2.j_2.j_3.j_1.j_4^{-2} \lambda_{12}^{(2)}$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{3,i_1,r_3} L_{a_2,i_2} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{1,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+3\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} L_{a_2,i_2} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1} \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} L_{a_2,i_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_2,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
cont.								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{16M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
cont.								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{4\epsilon M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8M_F} T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I} T 3.a_1.a_2.j_2.j_1 S_{1,j_1}^\dagger$	
								$+\frac{9\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8\epsilon M_F} T C P_L L_{a_1,i_1} g_S^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8M_F} T C P_L L_{a_1,i_1} g_S^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	
								$+3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	$+\frac{3\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{8M_F} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	
cont.								$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{\epsilon M_F} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_2} \overline{y}_{2,i_3,r_1} L_{a_2,i_2}}{M_F} T C P_L L_{a_1,i_1} g_Y^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
cont.									$+ \frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+ \frac{\sqrt{3}h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{\mathcal{I}T3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+ \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$- \frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_2}} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2,j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$- \frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2,j_2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}}$	$+ \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger}{M_F}$	$- 2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}}{\sqrt{3}}$
								$- 2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}2,j_2,j_1}^\dagger \lambda_{12}^{(1)}$		$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\mathcal{I}H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}}}{\sqrt{3}}$
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{\sqrt{3}M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
									$- \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger$	
cont.									$+ \frac{\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$	
									$+ \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$+ \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_2}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_3,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{2\sqrt{3}M_F}$	
									$- \frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+ \frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{1,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{y_{2,i_1,r_2}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
cont.									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$+ \frac{3h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{8M_F}$	
									$- \frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}\overline{y_{2,i_3,r_1}}\overline{L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$- \frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{\mathcal{I}T3,a_1,a_2,j_1} S_2^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T}CPL_{a_1,i_1}S_2\mathcal{IT}3_{a_1,a_2,j_1}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T}CPL_{a_1,i_1}S_2\mathcal{IT}3_{a_1,a_2,j_1}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0						$-\frac{4h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$			
													$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}S_2S_{1,j_2}y_{1,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}}\lambda_{12}^{(1)}}{\sqrt{3}}$			
													$-2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}S_2S_{1,j_2}y_{2,i_1,r_1}S_2^\dagger S_{1,j_1}^\dagger\mathcal{I}_{12,j_2,j_1}}\lambda_{12}^{(1)}$			
cont.													$-\frac{3h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger g_Y^2}{4\epsilon M_F}$			
													$+\frac{h\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger g_Y^2}{4M_F}$			
													$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T}CPL_{a_1,i_1}S_2\mathcal{IT}3_{a_1,a_2,j_1}S_{1,j_1}^\dagger g_Y^2}{4\epsilon M_F}$			
cont.													$+\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^T}CPL_{a_1,i_1}S_2\mathcal{IT}3_{a_1,a_2,j_1}S_{1,j_1}^\dagger g_Y^2}{4M_F}$			
													$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{8M_F}$			
cont.													$+\frac{\overline{L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{8M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{2\sqrt{3}M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{2M_F}$			
													$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}CPR\overline{L_{a_2,i_2}^T}S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{IT}3_{a_1,a_2,j_1}^*S_2^\dagger}{4\epsilon M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,i_2}^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,i_2}^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}$	$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}}{\sqrt{3}}$
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{\sqrt{3}\epsilon M_F}{2\sqrt{3}M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{2M_F}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_1} \overline{y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{M_F}$	$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	
cont.								$-\frac{\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{2M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \overline{y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger}}{2M_F}$	
cont.								$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger} \lambda_{12}^{(1)}}{M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	$-\frac{h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{4h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_2^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$	$-\frac{\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,i_2}^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}}{\sqrt{3}}$
								$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}}{\sqrt{3}}$	$-\frac{3\sqrt{3} h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1} S_{1,j_1}^\dagger S_{2,i_2}^\dagger}}{8M_F}$	$-2i\sqrt{\frac{2}{3}} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}$
								$-2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_2 S_{1,j_2} y_{2,i_1,r_1} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_2,j_1}^1 \lambda_{12}^{(1)}}$	$-\frac{9h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} g_{\tilde{S}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^{(2)} \lambda_{12}^{(2)}}}{\sqrt{3}}$
cont.								$-\frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} g_{\tilde{S}}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4M_F}$	$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	
cont.								$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{4\epsilon M_F}$	$-\frac{3h \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{8M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{\sqrt{3}\epsilon M_F}$	$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{\sqrt{3}\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	$-\frac{h \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_1} \overline{y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_2^\dagger}}{2\sqrt{3}M_F}$	

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.								$-\frac{2h\log\left(\frac{\hat{\rho}^2}{M_F}\right)}{\sqrt{3}}\frac{\overline{y_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2}}^{(3)}\wedge_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{\epsilon M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\rho}^2}{M_F}\right)}{\sqrt{3}}\frac{\overline{y_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2}}^{(3)}\wedge_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{\sqrt{3}M_F}$ $+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{\sqrt{3}M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{6M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_2,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{6M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{\sqrt{\frac{8}{3}}h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{\epsilon M_F}$ $-\frac{\sqrt{\frac{8}{3}}h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{8}{3}}h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{\sqrt{2}h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\epsilon M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\sqrt{2}M_F}$	
cont.									$-\frac{\sqrt{2}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\sqrt{2}M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}I\tau_{3a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{8M_F}$ $+\frac{\overline{y_{1,i_1,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$ $+\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\epsilon M_F}$	
									$-\frac{2h\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$ $+\frac{\overline{L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$	
cont.									$+\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$+\frac{3h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R \overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}hL_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_1}^\dagger$	$-\frac{\sqrt{6}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
								$+\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger g_2^2}{M_F}$	$-2i\left(\sqrt{\frac{\hat{\rho}^2}{M_F^2}}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger$
								$-\frac{4}{3}h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$-\frac{2\sqrt{3}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\hat{a}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}$	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{\sqrt{6}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T_{3,a_1,a_2,j_2,j_1}S_{2,j_1}^\dagger g_2^2}{M_F}$	
cont.								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{2\sqrt{3}h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\epsilon M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{2h\overline{L}_{a_1,i_1}CPR\overline{L}_{a_2,i_2}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger g_2^2}{\sqrt{3}M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\epsilon M_F}$	
cont.								$-\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{6}M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$	
								$2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}^\dagger 2\lambda_{12}^{(2)}$	$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T_{3,a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{4\sqrt{2} M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{36 M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_1,j_3,j_4,j_2} {}^\lambda \Lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{3\sqrt{2} M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^3 \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{2\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^3 \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_2,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{4\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1} {}^\dagger S_{2,j_2} {}^\dagger I_{12,j_3,j_1,j_4,j_2} {}^3 \Lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{2\sqrt{2} M_F}$	
cont.									$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{4\sqrt{2} M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{2\sqrt{2} M_F}$	
									$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{2\sqrt{2} M_F}$	
cont.									$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{4\sqrt{2} M_F}$	
									$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2} S_{1,j_1} {}^\dagger}}{3\sqrt{2} M_F}$	
cont.									$+\frac{\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} {}^T C P_L L_{a_1,i_1} g_V^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{4\sqrt{3} M_F}$	
									$+\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} {}^T C P_L L_{a_1,i_1} g_V^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{4\sqrt{3} M_F}$	
									$+\frac{\overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{\sqrt{3} M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{8 M_F}$	
									$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{8 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{46 M_F}$	
									$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{8 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{8 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{46 M_F}$	
									$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{8 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{46 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{6 M_F}$	
									$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2} {}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1} {}^\dagger}{36 M_F}$	
cont.									$-\frac{h$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
									$+ \frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_3,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	
cont.									$+ \frac{\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$ $+ \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\sqrt{3}M_F}$ $+ \frac{\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{\sqrt{3}M_F}$	
cont.									$+ \frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$ $+ \frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_3,r_3} y_{2,i_2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3\epsilon M_F}$	
cont.									$- \frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{6M_F}$ $- \frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3\epsilon M_F}$ $- \frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{6M_F}$	
cont.									$- \frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_3,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$ $- \frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_3} y_{1,i_2,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $- \frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$- \frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+ \frac{\sqrt{\frac{2}{3}} h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$ $+ \frac{\sqrt{\frac{2}{3}} h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$- \frac{h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $- \frac{h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$ $+ \frac{\sqrt{\frac{2}{3}} h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T^{3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+ \frac{\sqrt{\frac{2}{3}} h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T^{3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$ $+ \frac{3\sqrt{\frac{2}{11}} h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\epsilon M_F}$	
cont.									$- \frac{3\sqrt{\frac{2}{11}} h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\epsilon M_F}$ $- \frac{h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$ $- \frac{h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$- \frac{h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$ $+ \frac{3\sqrt{\frac{2}{11}} h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T^{3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\epsilon M_F}$ $+ \frac{3\sqrt{\frac{2}{11}} h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T^{3,a_1,a_2,j_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$	
cont.									$- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $- \frac{h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$ $- \frac{7h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$- \frac{7h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$ $- \frac{7h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $- \frac{7h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
cont.									$- \frac{7h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $- \frac{7h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $+ \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $+ \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$	$- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $- \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_1}^\dagger$ $+ \frac{4}{3} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$
									$- \frac{6h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} g_S^2 S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{\epsilon M_F}$ $- \frac{\sqrt{\frac{2}{3}} h\overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} g_S^2 S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{M_F}$ $+ \frac{\sqrt{\frac{2}{3}} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L}_{a_2,i_2}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T^{3,a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger}{M_F}$	$- \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $- 2i\sqrt{\frac{2}{3}} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,a_1,a_2,j_2,j_1}^\dagger$ $- \frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\epsilon M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\sqrt{2} M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{2\sqrt{2} \epsilon M_F}$	
cont.								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{4\sqrt{2} M_F}$	
								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\epsilon M_F}$	
								$+\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_1,j_2,j_3,j_4} \lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger}{3\sqrt{2} M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2\sqrt{3} h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} y_{1,i_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\epsilon M_F}$	
cont.								$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger I_{12,j_3,j_1,j_4,j_2} \lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} y_{1,i_2}^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$	
								$+\frac{\overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_2}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{6 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} \overline{y_{2,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$+\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\frac{h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{3\epsilon M_F}$	$-\frac{3h \overline{y_{1,i_1,r_1}} y_{2,i_2,r_2} \overline{y_{2,i_3,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8 M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$-\frac{\sqrt{2}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{3\sqrt{2}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$			
cont.													$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger}{4\sqrt{2}\epsilon M_F}$			
													$+\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
cont.													$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\epsilon M_F}$			
													$-\frac{2h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^Tg_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
													$+\frac{\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{\sqrt{3}M_F}$			
cont.													$+\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{6M_F}$			
cont.													$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger}{8M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tau_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tau_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
cont.													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
cont.													$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tau_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$			
													$-\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
cont.													$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$			
													$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$			
cont.													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$			
cont.													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
													$-\frac{7h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_2,i_2}}^TCPl_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F$	$S_1 S_2 \ell$	$S_1 F H$
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}\overline{L}_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \overline{y}_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{6M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{4\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{\sqrt{6}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,i_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{\epsilon M_F}$	
									$-\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{M_F}$	
									$-3\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger$	
cont.									$-\frac{\sqrt{\frac{2}{3}}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{2M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{4\sqrt{2}\epsilon M_F}$	
									$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{3\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{4\sqrt{2}\epsilon M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} \overline{y}_{2,i_2,r_2} y_{1,i_3,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{3h\overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_2} y_{1,i_1,r_2} \overline{y}_{2,i_2,r_2} y_{1,i_3,r_2} \overline{I} T \overline{a}_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger}{4\sqrt{2}\epsilon M_F}$	
									$+\frac{h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$	
									$+\frac{3h\overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} \overline{y}_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$	
cont.									$-\frac{2\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{\epsilon M_F}$	
									$-\frac{2h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{3\sqrt{3}h\overline{L}_{a_1,i_1} C P_R \overline{L}_{a_2,i_2}^T g_Y^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
cont.													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
													$-3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$ $-3\sqrt{\frac{2}{11}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
cont.													$-3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$ $-3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_2,j_1}^*S_{2,j_1}^\dagger\lambda_{12}^{(2)}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
													$-7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$ $-7h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger\lambda_{12}^{(3)}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
													$-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$ $-\frac{7h\overline{L_{a_1,i_1}}CPr\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
cont.													$-2\sqrt{6}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger$ $-2\sqrt{6}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger$ $-\frac{2\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$		$-2i\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\frac{\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}}$	
													$-\frac{2\sqrt{6}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{\epsilon M_F}$ $-\frac{2\sqrt{6}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}Y_{e,i_2,i_3}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}H_{1,a_1,a_2,j_2,j_1}^\dagger}{\sqrt{3}}$	
													$-\frac{2\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}{}^{2\lambda_{12}^{(1)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_3,j_1,j_4,j_2}{}^{2\lambda_{12}^{(1)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
cont.													$-\frac{2\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{\sqrt{2}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{3\sqrt{2}M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_2,j_3,j_4}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
													$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{4\sqrt{2}M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$		$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$ $+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}T_{12,j_1,j_3,j_4,j_2}{}^{2\lambda_{12}^{(2)}}}{\sqrt{3}}$	
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}T\tilde{\mathfrak{a}}_{a_1,a_2,j_1,j_2}S_{1,j_1}^\dagger}{2\sqrt{2}\epsilon M_F}$ $-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}$			

[illegible]

[illegible]

Field combinations 8 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	ℓ	$S_2 F$	ℓ
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$				
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0				
T3-A	0	1	0	3	1	2	$\frac{1}{2}$				
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$				
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0				
cont.											
cont.											
cont.											
T3-B	0	2	0	2	1	1	$\frac{1}{2}$				

[illegible]

[illegible]

Field combinations 9 of 10

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	S_1	S_2	F	ℓ
cont.															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_1} L_{a_4,1,2}{}^T C P_L L_{a_1,1,1} S_{2,j_1} y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4M_F}$			
											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_2 y_{2,1,3},r_1} L_{a_2,1,2}{}^T C P_L L_{a_1,1,1} S_{2,j_1} y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_1 y_{2,1,3},r_2} L_{a_2,1,2}{}^T C P_L L_{a_1,1,1} S_{2,j_1} y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_1^\dagger}{\sqrt{3}M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3}M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3}M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$			
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_1} L_{a_4,1,2}{}^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_3,1,1} C P_R L_{a_4,1,2}{}^T y_{1,1,1},r_1 y_{2,1,2},r_1 \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$			
											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{1,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_1} L_{a_4,1,2}{}^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_3,1,1} C P_R L_{a_4,1,2}{}^T y_{1,1,1},r_1 y_{2,1,2},r_1 \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$			
											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{1,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1 y_{2,1,2},r_1} L_{a_4,1,2}{}^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_3,1,1} C P_R L_{a_4,1,2}{}^T y_{1,1,1},r_1 y_{2,1,2},r_1 \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$			
											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} \lambda_{T3} S_{2,j_1} y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1^\dagger}{\sqrt{3}}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{1,1,1},r_1 \mathcal{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_1 y_{2,1,1},r_1 \mathcal{I} T_{3,a_1,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
cont.											$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$				$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{2,1,3},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			
															$h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\overline{y_{1,1,1},r_1} L_{a_1,1,1} C P_R L_{a_2,1,2} S_1 y_{1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_2 \mathcal{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H
											S_1	S_2	F	H
											$\sqrt{3}h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,3},r_2 y_{2,1,1},r_2 y_{2,1,2},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
cont.											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{1,1,3},r_2 y_{2,1,2},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											M_F			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{2,1,2},r_2 y_{2,1,3},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$\sqrt{3}M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{1,1,2},r_1 y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
cont.											$\sqrt{3}h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,3},r_1} \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_2 y_{2,1,2},r_1 y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_1} \overline{I_{T3,a_1,a_2,j_1}} S_{1,1}^\dagger} \lambda_{12}^{(1)}$			
cont.											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,1,1}} C P_R \overline{L_{a_2,1,2}} \overline{S_{1,y_1,1,1},r_1} \overline{y_{2,1,2},r_1} \overline{I_{T3,a_1,a_2,j_1}} S_{2,j_1}^\dagger \lambda_{12}^{(1)}$			
											M_F			
T3-B	-2	2	-1	2	0	1	-1/2				$+\frac{2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{L_{a_4,1,2}} \overline{C P_L L_{a_2,1,1}} H_{a_1} H_{a_3} \lambda_{T3,a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$			
											$2h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,1,1}} C P_R \overline{L_{a_4,1,2}} \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{L_{a_1,a_3} \epsilon_{a_2,a_4}} H_{a_1}^\dagger H_{a_2}^\dagger$			
											$+\frac{\sqrt{3}M_F}{\sqrt{3}M_F}$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{\lambda_{T3} S_{2,j_2} y_{1,1,1},r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,1,1},r_1} \overline{I_{T3,a_1,a_2,j_1,j_2}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$			
											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,1,1},r_1 \overline{I_{T3,a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger$			
cont.											$-\frac{1}{2} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,1,1},r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,1,1},r_1 \overline{I_{T3,a_1,a_2,j_2,j_1}} \lambda_{T3} S_{2,j_1}^\dagger$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{1,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_2} \overline{I_{T3,a_1,a_2,j_1}} \overline{I_{T2,a_2,j_2}} S_{1,j_1}^\dagger$			
											$4M_F$			
											$h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,1,1},r_1} \overline{y_{2,1,2},r_2} \overline{y_{2,1,3},r_2} \overline{L_{a_2,1,2}} \overline{C P_L L_{a_1,1} S_{2,j_2} y_{2,1,3},r_$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$	
cont.									$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} H_{a_1}^\dagger \lambda_{H2}^{(1)}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1}^* L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	
									$+h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{1,a_2},j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	
									$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{M_F}$	
									$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{y_{1,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	
cont.									$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{cg}^1_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	
									$- \frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{1,a_2},j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	
cont.										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* g^2_{S_1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$
cont.										$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1}^* L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \epsilon_{a_1,a_2} S_{2,i_1}^\dagger \lambda_{H2}^{(1)}}}{M_F}$
										$+\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1}^* L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1} L_{a_2,i_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1}^* L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{\sqrt{3} M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{H2}^{(1)}}}{\sqrt{3} M_F}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2} \mathcal{I}_{a_4}^* C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{\sqrt{3} M_F}$	$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$	$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{a_2}^* L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} g^2_{S_2,j_2} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	
								$+\frac{2h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2} \mathcal{I}_{a_4}^* y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}}{\sqrt{3} M_F}$	$+\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_3} y_{2,i_2,r_2} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	
								$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	
									$-\frac{1}{2} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_2,r_2} y_{2,i_3,r_3} L_{a_2} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_3} \mathcal{I}_{a_1}^* \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} g^2_{S_2,j_2} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{a_2}^* C P_L L_{a_1,i_1} g^2_{S_2,j_2} \mathcal{I}_{y_{2,a_1},j_1} \mathcal{I}_{y_{2,a_2},j_2} S_{1,j_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_3,r_3} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} y_{2,i_2,r_2} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$
										$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{2M_F}$	$-\frac{h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2} \mathcal{I}_{a_2}^* S_{1,j_2} y_{1,i_1,r_1} y_{2,i_3,r_3} \mathcal{I}_{a_1} \mathcal{I}_{a_1,i_1} \mathcal{I}_{y_{2,a_2},j_2} S_{2,i_1}^\dagger}}{4M_F}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_3}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{x_{a_1,i_1},j_1} I_{x_{a_2,i_2},j_2} S_{1,j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{x_{a_1,i_1},j_1} I_{x_{a_2,i_2},j_2} S_{1,j_1}^\dagger}{2 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^{(2)} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{4 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} g_{S_1,j_2}^{(2)} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$
									$+\frac{6 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{e_{a_1,\omega_1}} C P_R \overline{L_{a_2,i_2}} g_{S_1,j_2}^{(2)} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{2 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_2,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_3,r_2} y_{2,i_2,r_1} y_{1,i_2,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{2 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger}{4 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{I_{a_1,\omega_1}} C P_R \overline{L_{a_2,i_2}} g_{S_1,j_1}^{(2)} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,\omega_2} S_{2,j_1}^\dagger}{4 M_F}$	$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} e_{a_1,\omega_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
									$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,\omega_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} I_{x_{a_1,\omega_1},j_1} I_{x_{a_2,\omega_2},j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} I_{x_{a_1,\omega_1},j_1} I_{x_{a_2,\omega_2},j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} I_{x_{a_1,\omega_1},j_2} I_{x_{a_2,\omega_2},j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\begin{aligned} & -\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} e_{a_1,\omega_2} e_{a_3,\omega_4}}{3 M_F} \\ & -\frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{a_1,\omega_2} e_{a_2,\omega_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3 M_F} \end{aligned}$	$\begin{aligned} & -\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} I_{T3,a_1,\omega_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} \\ & -\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} I_{T3,a_1,\omega_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger} \\ & -\frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} I_{T3,\omega_1,\omega_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger \end{aligned}$	
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,\omega_1,\omega_2,j_2,j_1} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{9 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^{(2)} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
cont.									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$	$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{a_1,\omega_2} S_{1,j_1}^\dagger}{8 M_F}$
									$+\frac{3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	S_1S_2FH	$S_1S_2F\ell$
cont.										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{8M_F}$
										$+\frac{9\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
cont.										$-\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{4M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{4M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}}{M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}}{M_F}$
										$+\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} I T 3_{a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}}{\sqrt{3}M_F}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}}{3M_F}$	$-2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} \text{Bar}(\text{eg}4)_{a_2,j_1,a_1,j_2} H_{a_1}{}^\dagger}$
									$-\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1} C P_R L_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}}{3M_F}$	$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} I T 3_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}$
cont.										$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} I T 3_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}$
										$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} I T 3_{a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$
cont.										$-\frac{1}{2}\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} I T 3_{a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}{}^\dagger}$
										$-2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3} \text{eg}4_{a_1,j_2,a_2,j_1} H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}$
cont.										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{eg}3)_{j_1,j_2} H_{a_1}{}^\dagger \lambda_{H2}^{(1)}}$
										$-3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \text{Bar}(\text{eg}4)_{a_2,j_1,a_1,j_2} H_{a_1}{}^\dagger \lambda_{H2}^{(1)}}$
cont.										$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{eg}3_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{H2}^{(1)}}$
										$-3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{eg}4_{a_1,j_2,a_2,j_1} H_{a_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{H2}^{(1)}}$
cont.										$+\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{eg}4)_{a_2,j_1,a_1,j_2} H_{a_1}{}^\dagger \lambda_{H2}^{(2)}}$
										$+\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \text{eg}4_{a_1,j_1,a_2,j_2} H_{a_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{H2}^{(2)}}$
cont.										$-\frac{3\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{4M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{8M_F}$
										$+\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}}{4M_F}$
cont.										$+\frac{3\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{8M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{8M_F}$
cont.										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{4M_F}$
										$+\frac{\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1} C P_R L_{a_2,i_2}{}^T S_{1,j_2} y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} I T 3_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}}{2M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
										$\begin{aligned} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{8 M_F} \\ \text{cont.} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ \text{cont.} & 3 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} g_2^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & - \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & - \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{4 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{4 M_F}{4 M_F} \\ \text{cont.} & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \\ & + \frac{8 M_F}{8 M_F} \\ \text{cont.} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)} \\ & + \frac{M_F}{M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)} \\ & + \frac{M_F}{M_F} \\ & h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} T r s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}} \\ & + \frac{\sqrt{3} M_F}{\sqrt{3} M_F} \\ & h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{\alpha_1,i_1}} C P_R \overline{L_{\alpha_2,i_2}^\dagger} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} T r s_{\alpha_1,\alpha_2,j_1,j_2}^T S_{2,j_1}^\dagger \lambda_{12}^{(2)} \\ & + \frac{\sqrt{3} M_F}{\sqrt{3} M_F} \end{aligned}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & - \frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_4,i_2}{}^T C P_L L_{\alpha_2,i_1} H_{\alpha_1} H_{\alpha_3} \lambda_{T3} e_{\alpha_1,\alpha_2} e_{\alpha_3,\alpha_4}}} {3 M_F} \\ & - \frac{2 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{\alpha_3,i_1}} C P_R \overline{L_{\alpha_4,i_2}^\dagger} y_{1,i_1,r_1} y_{2,i_2,r_1} e_{\alpha_1,\alpha_3} e_{\alpha_2,\alpha_4} H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger} {3 M_F} \\ \text{cont.} & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_1,j_2}^*} H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger S_{1,j_1}^\dagger} \\ & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \overline{\lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_1,j_2}^*} H_{\alpha_1}^\dagger H_{\alpha_2}^\dagger S_{1,j_1}^\dagger} \\ & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{\alpha_1} H_{\alpha_2} S_{1,j_2} y_{2,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger} \\ & - \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{\alpha_1} H_{\alpha_2} S_{1,j_2} y_{2,i_1,r_1} I T s_{\alpha_1,\alpha_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger} \\ \text{cont.} & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} g_2^T S_{2,j_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger} \\ & + \frac{4 M_F}{4 M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_2} y_{2,i_3,r_2} T r s_{\alpha_1,\alpha_2,j_2,j_1} S_{1,j_1}^\dagger} \\ & + \frac{4 M_F}{4 M_F} \\ & 9 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} g_2^T S_{2,j_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{1,i_3,r_1} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{4 M_F}{4 M_F} \\ \text{cont.} & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{1,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{4 M_F}{4 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{4 M_F}{4 M_F} \\ \text{cont.} & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha_2,i_2}{}^T C P_L L_{\alpha_1,i_1} S_{2,j_1} y_{2,i_3,r_2} e_{\alpha_1,\alpha_2} S_{1,j_1}^\dagger} \\ & + \frac{8 M_F}{8 M_F} \\ & 3 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{\alpha$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} W_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} W_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} W_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
cont.										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} W_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{MF}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{MF}$
cont.										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} MF}$
										$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} MF}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$- \frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} \lambda T_{3,\epsilon_{a_1},a_2,\epsilon_{a_3},a_4}}{3MF}$	$- \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{\lambda T_{3,S_2,j_2}} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$- \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
								$- \frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda T_{3,L a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{3MF}$	$- \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} \overline{\lambda T_{3,S_2,j_2}} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+ \frac{9h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
									$- \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda T_{3,S_2,j_1}^\dagger$	$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
cont.									$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{4MF}$	$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
									$- \frac{1}{2} \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda T_{3,S_2,j_1}^\dagger$	$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} y_{2,i_3,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} y_{2,i_3,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{8MF}$
cont.										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} y_{2,i_3,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger}{4MF}$
										$+ \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 S_{2,j_1}^\dagger \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
										$+ \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 S_{2,j_1}^\dagger \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
cont.										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
cont.										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{8MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger}{4MF}$
cont.										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
										$- \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{9h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1}^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
cont.										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{4MF}$
										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_3,r_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
cont.										$+ \frac{\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
										$- \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
cont.										$+ \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$
										$+ \frac{9\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{S_1}^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8MF}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
$cont.$										$- \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$ $+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
$cont.$										$+ \frac{3h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{8M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $+ \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
$cont.$										$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2} \overline{I} T_{3,a_1,a_2,j_2,j_1} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$ $+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{I} T_{3,a_1,a_2,j_1,j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$
$T3-D$	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}^T} C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+ \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}^T} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3} M_F}$	$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger}{\sqrt{3}}$ $- h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \lambda_{T3} S_{2,j_1,i_1,r_1} \overline{I} T_{3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_1,r_3} \epsilon_{a_1,a_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{\sqrt{3} M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_1,r_3} \epsilon_{a_1,a_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}^T} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{2,j_1}^\dagger}{4M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_1} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{\sqrt{3} M_F}$ $- \frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T} C P_L L_{a_1,i_1} S_{2,j_2,i_3,r_2} \overline{I} T_{3,a_1,a_2,j_1} S_{1,j_1}^\dagger}{4M_F}$ $- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{\sqrt{3}M_F}$	$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
									$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
									$\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{M_F}$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L}_{a_1,i_1}C P_R \overline{x}_{a_2,i_2}^\dagger S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_2,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger\lambda_{12}^2}{M_F}$
cont.									$\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{\lambda}_{T3}S_2 y_{1,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$	$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_1,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda}_{T3}L_{a_3,i_1}C P_R \overline{L}_{a_4,i_2}{}^r y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{\sqrt{3}M_F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}}$	$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_1,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\lambda_{T3}L_{a_3,i_1}C P_R \overline{L}_{a_4,i_2}{}^r y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{\lambda}_{T3}S_2 y_{1,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}\overline{\lambda}_{T3}S_2 y_{2,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{M_F}$
										$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{\sqrt{3}M_F}$
										$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
cont.									$\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_2,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}e_{a_1,a_2}e_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger}{\sqrt{3}}$	$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$	$\frac{-9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r g_F^2 S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda}_{T3}L_{a_3,i_1}C P_R \overline{L}_{a_4,i_2}{}^r y_{1,i_1,r_1}y_{2,i_2,r_1}e_{a_1,a_3}e_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_1,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{\lambda}_{T3}S_2 y_{1,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{\sqrt{3}M_F}$
										$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_1,r_1}\overline{\lambda}_{T3}S_2 y_{2,i_1,r_1}\overline{x}_{T3,a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}L_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_1,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{2,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{M_F}$
										$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}\overline{L}_{a_1,i_1}C P_R \overline{L}_{a_2,i_2}{}^r S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_1,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_2^\dagger}{4M_F}$
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger$	$\frac{-9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_F^2 S_2\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
										$\frac{-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$
										$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{\sqrt{3}M_F}$
cont.									$\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{1,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$	$\frac{-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_2 y_{2,i_3,r_2}\overline{x}_{T3,a_1,a_2,j_1}S_{1,j_1}^\dagger}{4M_F}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
										$-\frac{\sqrt{6} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}} \ell^\dagger g_2^* S_{1, j_2} y_{i_1, r_1} y_{i_2, r_1} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger}{M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_3, r_2} y_{i_2, r_1, r_2} y_{i_3, r_1} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger}{2 \sqrt{2} M_F}$ $+ \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_1, r_2} y_{i_3, r_1} y_{i_2, r_2} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger}{3 M_F}$ $- \frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_1, r_2} y_{i_2, r_2} y_{i_3, r_1} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger}{3 M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_1, r_2} y_{i_1, i_2, r_1} y_{i_2, i_3, r_2} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger}{2 \sqrt{2} M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_2, r_2} y_{i_1, i_3, r_2} y_{i_2, i_2, r_1, r_2} e_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$ $+ \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_2, r_2} y_{i_1, i_3, r_2} y_{i_2, i_2, r_1, r_2} e_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$ $+ \frac{2 \sqrt{3} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger g_2^* S_{1, j_1} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} e_{a_1, a_2} S_{2, j_1}^\dagger}{M_F}$ $+ \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_3, r_2} y_{2, i_2, r_1, r_2} e_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_1, r_2} y_{i_1, i_3, r_2} e_{a_1, a_2} S_{2, j_1}^\dagger}{3 M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_1, r_2} y_{2, i_2, r_2, r_1} e_{a_1, a_2} S_{2, j_1}^\dagger}{3 M_F}$ $+ \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_3, r_1}} L_{a_1, i_1}^\dagger C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_1} y_{i_1, i_1, r_2} y_{2, i_2, r_2} e_{a_1, a_2} S_{2, j_1}^\dagger}{4 M_F}$ $+ \frac{\sqrt{8} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_1, i_1, r_1}} y_{i_2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} \bar{T} \tau_{a_1, a_2, j_1, j_2} S_{i, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_1, i_1, r_1}} y_{i_2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} e_{a_1, a_2} S_{i, j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$ $+ \frac{\sqrt{8} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{i, j_1} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} e_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$ $+ \frac{3 \sqrt{\frac{3}{11}} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_1, i_1, r_1}} y_{i_2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_2} \bar{T} \tau_{a_1, a_2, j_1, j_2} S_{i, j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_1, i_1, r_1}} y_{i_2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} e_{a_1, a_2} S_{i, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$ $+ \frac{3 \sqrt{\frac{3}{11}} h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{1, j_2} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} \bar{T} \tau_{a_1, a_2, j_2, j_1}^* S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$ $- \frac{h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{i, j_1} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} e_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$ $+ \frac{7 h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{y_{i_1, i_1, r_1}} y_{i_2, i_2, r_1} L_{a_2, i_2}^\dagger C P_L L_{a_1, i_1} S_{2, j_1} e_{a_1, a_2} S_{i, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$ $+ \frac{7 h \log \left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1, i_1}} C P_R \overline{L_{a_2, i_2}}^\dagger S_{i, j_1} y_{i_1, i_1, r_1} y_{i_2, i_2, r_1} e_{a_1, a_2} S_{2, j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{1,1,3} y_{2,2,1} y_{2,2,2} \tau_2 \tau_2}^T C P_L L_{a_1,1} S_{2,j_1} y_{1,1,3,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{1,1,3} y_{2,2,1} y_{2,2,2} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{1,1,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{3 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{1,1,2} \tau_2 y_{2,2,3} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{1,1,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 y_{2,2,1} y_{2,2,2} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{2,2,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,2,2} \tau_2 y_{2,2,3} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{2,2,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,2,2} \tau_2 y_{2,2,3} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{2,2,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,2,2} \tau_2 y_{2,2,3} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{2,2,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,1} \tau_1 y_{2,2,2} \tau_2 y_{2,2,3} \tau_1}^T L_{a_2,1,2}^T C P_L L_{a_1,1} S_{2,j_1} y_{2,2,3,1} y_{2,2,1} \epsilon_{a_1, a_2} S_{1,j_1}^\dagger}{4 M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{3 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{2 \sqrt{2} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{2 M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{2 \sqrt{2} M_F}$
cont.										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{2 \sqrt{2} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\dagger}}{2 \sqrt{2} M_F}$
										$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,1,3} \tau_1 L_{a_1,1}^\dagger C P_R \overline{L_{a_2,1,2}^\dagger} S_{1,j_2} y_{1,1,2,1} y_{1,1,3,1} y_{2,2,1,1} y_{2,2,1,1} \tau_2 I T \tau_{a_1, a_2, j_2, j_1}^\dagger S_{2, j_1}^\d$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{2,1,1}^{\dagger} H_{a_1} H_{a_2} S_{1,j_2} Y_{2,1,1}^{\dagger} \mathcal{I} T_{3,a_1,a_2,j_2,j_1} \lambda T_{3,2} S_{2,j_1}^{\dagger}}}{\sqrt{3}}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{1,i_2,r_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} Y_{1,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^{\dagger}}}{2\sqrt{2} M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} Y_{2,i_2,r_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} Y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^{\dagger}}}{2\sqrt{2} M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} Y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^{\dagger}}}{2\sqrt{2} M_F}$
cont.									$-\frac{\sqrt{2} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} Y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} Y_{2,i_3,r_2} \mathcal{I} T_{3,a_1,a_2,j_1,j_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{2\sqrt{3} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_{21}^2 S_{2,j_1}^{\dagger} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{1,i_3,r_2} Y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{1,i_3,r_2} Y_{2,i_2,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} Y_{1,i_2,r_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} Y_{1,i_2,r_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{4 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
cont.									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$	$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger}}}{3 M_F}$
									$-\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,i_1} S_{2,j_2} Y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}^{\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	ℓ
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				$\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{M_F}$ $\frac{2\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}}L_{a_3,i_1}C\overline{P_R}L_{a_4,i_2}{}^t y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{1,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{\lambda_{T3}}S_{2,j_2}y_{2,i_1,r_1}I_{T3}a_{a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}I_{T3,a_1,a_2,j_2,j_1}\lambda_{T3}S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$	
cont.											$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{y_{2,i_2,r_2}}y_{1,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}I_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_2}}y_{1,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{3\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}g_{S_2,j_2}^2\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$		
cont.											$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$	$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$		
cont.											$-\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_1,j_2}^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_1,j_2}^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_3,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$	$-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$		
cont.											$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{2\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_1,j_2}^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{3\sqrt{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t g_{S_1,j_2}^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{M_F}$	$+\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$		
cont.											$-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_2}I_{T3,a_1,a_2,j_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\overline{P_L}L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	$+\frac{\sqrt{\frac{3}{2}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}I_{T3}a_{a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C\overline{P_R}L_{a_2,i_2}{}^t S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$		
cont.														

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell$	$S_1 S_2 F H$	$S_1 S_2 F \ell$
cont.									$\frac{3\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1}S_{2,j_1}}^{\dagger}\lambda_{12}^{(2)}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	$\frac{3\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPH\overline{L_{a_2,i_2}}^TS_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\overline{IT_{3,a_1,a_2,j_1}S_{2,j_1}}^{\dagger}\lambda_{12}^{(2)}}}{M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(2)}}}{\sqrt{11}M_F}$
									$\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^{\dagger}\lambda_{12}^{(3)}}{\sqrt{23}M_F}$ $-\frac{7h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{L_{a_1,i_1}CPR\overline{L_{a_2,i_2}}^TS_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}^{\dagger}\lambda_{12}^{(3)}}}{\sqrt{23}M_F}$	
cont.									$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_{T3}S_{2,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}}}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_{T3}S_{2,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_2^2S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1}S_{2,j_1}^{\dagger}}}{M_F}$ $-\frac{2\sqrt{6}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}g_1^2S_{2,j_2}\overline{IT_{3,a_1,a_2,j_1}S_{2,j_1}^{\dagger}}}{M_F}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}CPH\overline{L_{a_4,i_2}}^{\dagger}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}}}{M_F}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$ $-\frac{\sqrt{2}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{3M_F}$
cont.									$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{3M_F}$
cont.									$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{3M_F}$
cont.									$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\overline{IT_{3,a_1,a_2,j_1}\lambda_{T3}S_{2,j_1}^{\dagger}}}}{\sqrt{3}}$	$\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline{IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}}{2\sqrt{2}M_F}$ $-\frac{h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}^TCPL_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\overline$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ
										$S_1 S_2 F$
										$S_1 S_2 F \ell$
										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
cont.										$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$
										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_3,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
cont.										$+ \frac{\sqrt{\frac{3}{2}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$
cont.										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
										$3 \sqrt{\frac{2}{11}} h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1}^\dagger S_{2,j_1}^\dagger \lambda_{12}^{(2)}$
										$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
cont.										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$
										$7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}$
										$- \frac{7 h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
cont.								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell e$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			